

Parikshit Mahajan

M.Tech (Research), Computational and Data Sciences • CGPA: 8.30



ACADEMIC PROFILE

M.Tech (Research)	8.30	Indian Institute of Science (IISc), Bengaluru	2027
B.Tech, Production Engineering	8.13	Veeramata Jijabai Technological Institute (VJTI), Mumbai	2024
Class XII (Maharashtra Board)	76.00	Pratap College, Amalner, Jalgaon	2019
Class X (Maharashtra Board)	93.40	Sane Guruji Nutan Madhyamik Vidyalaya, Amalner	2017

COURSES

Pattern Recognition and Neural Networks	Stochastic Models and Applications
Advanced Deep Representation Learning	Estimation Theory
Optimization for Machine Learning	Real Analysis
Numerical Linear Algebra	Measure Theory and Integration
Numerical Solution of Differential Equations	Functional Analysis
Numerical Methods	Introduction to Scalable Systems

PROJECTS

AW-FNO: A Neural Operator for Parametric Partial Differential Equations (M.Tech (Res) Thesis)	Dec'25–Jul'26
- Designed and implemented AW-FNO, a hybrid deep learning model combining Fourier Neural Operator (FNO) and Wavelet Neural Operator (WNO) branches in a parallel architecture to predict solutions of parametric partial differential equations. - Developed an Adaptive Gated Fusion Mechanism, a learned gate that routes each point of the input to the better-performing branch, adding negligible parameters and remaining constant in cost as grid resolution grows. - Reduced mean relative L2 error by 45x over the FNO baseline on the Burgers benchmark and by 1.8x on Darcy flow, and achieved 57.40 dB PSNR at 4x super-resolution of turbulent flow fields. - Benchmarked the model against parameter-matched FNO and WNO baselines across four datasets using Python, PyTorch and GPU-accelerated training on an NVIDIA RTX A6000. Links: github.com/parikshit-airex-lab/AW-FNO ICCFD13 Proceedings (Paper 311)	
Retrieval-Augmented Generation: NumPy Retrieval Pipeline and LangGraph Transcript Assistant	Aug'26–Sep'26
- Built a RAG retrieval pipeline in NumPy around a pretrained sentence encoder: overlapping character chunking, normalised embeddings and cosine similarity search, without any retrieval framework. - Built a transcript question-answering assistant using LangChain, LangGraph and Streamlit, with a Chroma vector store and a Groq-hosted LLM. - Structured the assistant as a two-node retrieve-and-generate graph with SQLite checkpointing, giving resumable chat threads across sessions. - Benchmarked the two implementations on a shared question set to compare retrieval quality and latency.	
Deep Learning Models from First Principles: A Unified Training Framework	Sep'26–Present
- Implemented CNN, RNN, LSTM and Transformer models from first principles in PyTorch behind a single interface, with a shared training loop, config-driven runs and a registry that selects a model by name. - Built the surrounding framework layer once for all models: dataset loaders with a common batch format, checkpointing and resumable runs, metric logging and seeded, reproducible experiments.	

PUBLICATIONS AND TALKS

- P. Mahajan**^{*}, M. Saini^{*}, G. M. Jyrwa, D. Nagchaudhury. *AW-FNO: An Adaptive Approach for Fluid Flow Super-Resolution via Gated Wavelet-Fourier Learning*. **ICCFD13**, Milan, Italy, July 2026 (paper ICCFD13-2026-311). ^{*}Equal contribution; **presenting author**. Awarded the **IISc GARP** grant for conference travel.
- Opening Speaker, AI/ML and Trustworthy Systems Roundtable Discussion, IISc Google Day**, Bengaluru, September 2026. Presented AW-FNO research on neural operators for parametric partial differential equations.

POSITIONS OF RESPONSIBILITY

- Organizing Committee Member**, CASML 2025, the International Conference on Applied AI and Scientific Machine Learning, AiREX Lab, IISc Bengaluru, December 2025. A four-day international conference held as an official pre-summit event of the AI Impact Summit 2026, with a pre-conference workshop in collaboration with Google.
- Managed the conference website and led poster preparation with a three-member team. Ran the registration desk, issued delegate ID cards, and acted as an on-site point of contact for participant queries across all four days.

SKILLS

- Programming:** Python, C++
- Tools & Libraries:** PyTorch, NumPy, Pandas, scikit-learn, Matplotlib, Git, Docker, $\text{\LaTeX}$
- Technical:** Machine Learning, Deep Learning, Optimization, Statistical Modelling, Numerical Methods, Scientific ML

ACADEMIC ACHIEVEMENTS

- GATE 2024: AIR 134** in Production and Industrial Engineering; **AIR 2495** in Mechanical Engineering.
- Qualified the **Homi Bhabha Balvaidyanik** Competition, 2016.
- State Rank 184**, Maharashtra Olympiad Movement, 2015.
- Secured ranks in the Maharashtra Talent Search Examination (MTSE) for three consecutive years, 2014–17.

EXTRACURRICULAR ACTIVITIES

- First Prize**, Pantomime, Balsahitya Sammelan, 2016.
- Represented the school cricket team at **district level**, 2014–15, and at the district-level chess competition, 2016.
- Second Prize**, Drama, annual cultural festival, Pratap College, Amalner, 2018.

INTERESTS AND LANGUAGES

Interests: Cricket, chess, plays and drama. **Languages:** English, Hindi, Marathi.